Starting as overdensities in the initial density fluctuations and evolving into the largest gravitationally bound structures in the Universe, galaxy clusters are a rich testbed of the cosmos. Their large masses, self-similarity, combination of both luminous and dark matter, and multi-wavelength observables make them ideal candidates for studying cosmology. Specifically, I am interested in using galaxy clusters to probe the fundamental nature of dark energy through understanding the growth of structure in the Universe.

In order to set competitive limits, several systematic effects must be studied and mitigated, including those arising from optical cluster selection, blending-induced biases in shear and redshift calibration, and mass-estimation errors caused by model misspecification. Over the next few years, I plan to perform early science measurements using clusters observed by Roman and Rubin while developing novel mitigation and calibration methods for future data releases. My previous experiences with clusters and shear calibration, along with membership of LSST DESC and the Rubin commissioning team equip me with the necessary tools to successfully carry out this reesarch.

Previous and Current Research *Blending* During my PhD, I focused on the weak lensing systematic of **unrecognized blends** which are objects (typically galaxies) that are mistakenly identified as a single entity due to severe overlap between sources (see Fig. 1 for an example). The repercussions of unrecognized blends via detection bias alone have shown shifts in cosmology parameters at the level of a few percent [3–5], and an early study including the impact on shape and flux measurements on cosmic shear found up to a 2σ shift in S_8 [6].

In 3x2pt cosmological analysis, expensive image simulations are used to quantify the biases unrecognized blends introduce into redshift and shear calibration. Directly identifying unrecognized blends at the catalog level is an appealing solution that I pursued during my PhD.

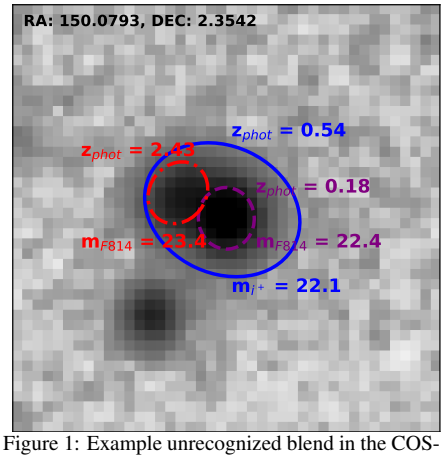


Figure 1: Example unrecognized blend in the COSMOS field with observed magnitudes and estimated photometric redshifts attached. Dark blue is the object as detected and measured by Subaru, dashed-dotted red and purple are detections and measurements from HST. From [7].

In [14], we showed that clustering based **machine learning techniques can identify unrecognized blends at the catalog level** by estimating the probability an object is an unrecognized blend based on measured fluxes alone. However, it was unclear if such a removal would ultimately improve our cosmology. As mentioned earlier, previous work suggested that removing unrecognized blends could improve cosmology, but those works make a number of assumptions that limit the applicability to modern cosmology. In order to decouple the effect of this removal on redshifts and shears, I investigated the efficacy of our blend removal method on shear calibration using the AnaCal shear framework. We showed that our blend removal scheme can not only reduce the bias, but also **totally eliminate the bias** with a 30-40% sample cut [7]. This is a promising mitigation strategy that can eliminate a substantial systematic bias in LSST weak lensing and **enable weak lensing analysis that use alternate tomographic bins** from 3x2 analysis like in cluster lensing or in dwarf galaxy lensing.

Clusters I led the LSST DESC Project 408 "A Rubin view of Abell 360", a collaborative effort between the Rubin Observatory and DESC, to study the galaxy cluster Abell 360 on commissioning data. This is the first weak lensing result to come from the Rubin Observatory and acts as

proof that the instrument is able to deliver science quality data even in suboptimal conditions. I contributed to **all steps of analysis** from red sequence identification to mass fitting My responsibilities included **source sample selection, photo-z validation, shear measurement and calibration with shapeHSM and AnaCal, cluster mass fitting**. I helped collaborators create aperture mass maps, study 1D galaxy distributions, and evaluate cuts with synthetic source injection (SSI). I also led scientific discussion, coordinated paper writing efforts, and organized almost daily telecons. This was my first foray into clusters along with leading a large collaborative effort, both of which were excellent learning experiences for me.

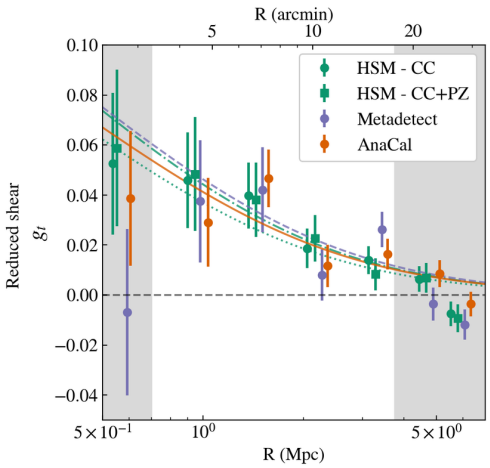


Figure 2: Tangential shear signal measured around Abell 360 using a variety of shape measurements and source selections. From [15]

Following my efforts on Abell 360, I have been testing the combination of the cutting-edge shear calibration, AnaCal, and photo-z estimators, RAIL, by measuring cluster profiles in the Euclid Deep Field South [publication in prep.] In this work we have **measured tangential shear signals around clusters that were undetectable in the base Rubin catalog** due to cleaner sample selection and higher quality shear estimates, highlighting the power of the combination of AnaCal and RAIL. The combination of these tools is currently being used for HSC Year 6 cosmic shear and **this is the first cluster analysis** using AnaCal+RAIL.

Throughout my PhD, I have worked extensively on weak lensing systematics and focused on measurement validation for LSST. I have developed new weak lensing mitigation strategies [7, 14], produced some of the first weak lensing results from Rubin [15], and have a familiarity with the tools used in modern cosmological analysis [16].

Research Proposal Galaxy clusters are powerful probes into the growth of structure in the Universe but can suffer from several sources of bias. Sub-percent level constraints on $\sigma_8(z)$ are achievable for LSST if the mass scatter can be constrained to within ±0.05 [17] whereas the latest cluster calibrations are limited to ~ 0.2 [18, 19]. The majority of that error budget comes from redshift calibration which is impacted by blending [18–20]. Furthermore, it is **currently unknown if modern shear calibrations, AnaCal and Metadetect, are able to meet the 1% shear calibration requirement needed for cluster weak lensing when accounting for realistic cluster conditions[srd]**. Accurately characterizing biases in cluster weak lensing is critical for achieving precision cosmology. These measurements will also guide the development of specialized tools for cluster weak-lensing analyses. Over the next few years, I propose to measure the level of shear bias in the cluster regime, work on novel methods of photo-z calibration, and produce cluster lensing profiles with Roman and Rubin.

Goal 1: Shear bias Precise ensemble measurements of galaxy clusters require an estimate of both statistical and systematic bias. For clusters, this is traditionally done with N-body or hydrodynamic simulations that can capture physics-induced bias (like miscentering of optical clusters due to triaxiality), but are computationally expensive and can ignore biases coming from image processing. Directly simulating co-adds is a substantially cheaper method that has been used to

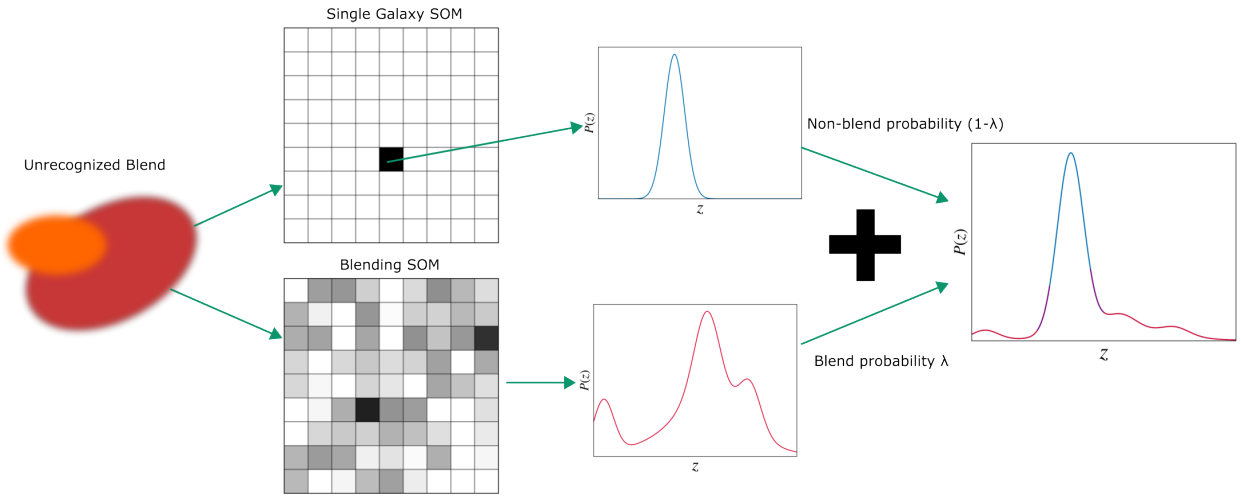


Figure 3: Example of modeling blends when producing redshift estimates via a SOM. Instead of only using the cell of the best fit, a weighted SOM can be made from synthetic blends that can be used to produce a more accurate PDF (color-coded to highlight contribution from individual terms).

calibrate biases in cosmic shear algorithms like AnaCal and Metadetect [8, 9, 12, 13].

These simulations are designed for cosmic shear, but their applicability to the clsuter regime — which experience higher shear and higer rates of blending — **has not been established**. I will test that assumption by measuring the shear bias for clusters using image simulation. Incorporating a redshift varying background population, foreground cluster members and contaminants, intra-cluster light, cluster triaxiality, and more realistic sky background variation into these simulations are some of the improvements needed to properly test shear calibration. Additionally, incorporating these systematics into our simulations sequentially will allow us to isolate the bias introduced by each effect and inform future mitigation strategies.

Goal 2: Photo-z calibration and joint processing As stated above, photo-z calibration, which is affected by blending and unrecognized blends, remains one of the largest hurdles to achieving sub-percent level constraints on $\sigma_8(z)$. My previous work on removing contaminated objects is a straightforward method that may improve the photo-z calibration at a statistical cost. An alternative approach is to use the machine learning probability from [14] as a prior and model certain objects a superposition of two galaxies. This may lead to more accurate posteriors for redshift without incurring a statistical cost.

The self-organizing map (SOM) is a powerful machine learning algorithm that projects the multidimensional color space onto a 2D surface. The same map that is used to create ensemble redshift estimates [10] can be extended to account for blends as outlined in Fig. 3. In that example, we use a blending probability (the prior from [14]) to combine two redshift probability distributions: the distribution assuming a single galaxy, and the distribution assuming a superposition of two galaxies. In preliminary work, I have shown that this schema can produce more accurate posterior estimates than a single-galaxy SOM approach. Accounting for more realistic blending, training-testing mismatch, and sample size variance are all necessary before applying this approach to real data.

The best solution to account for unrecognized blends would be to recognize them through joint processing of space and ground based imaging. While there are strong solutions on how to best

combine photometry, there is work to be done on combining shape measurements in order to maximize weak lensing signal. During my postdoc, I plan to contribute to the ongoing efforts to enable joint-processing for both Metadetect and AnaCal. This will mostly rely on simulated datasets, like the existing Roman-Rubin simulations, but can be tested on data using overlapping deep field observation of clusters which I am well equipped to study. If successful, this will completely circumvent the need for the machine learning techniques I have previously worked on wherever there is overlap between ground and space instrument. Regardless of the solution used, improving the photo-z calibration and reducing shape noise at higher redshifts is key to constraining the growth of structure with clusters.

Goal 3: Cluster measurements for Roman and Rubin With a validated pipeline in hand, we will have confident (or at least properly weighted) cluster weak lensing masses. Rubin Data Preview 2 includes $\sim$ 100 X-Ray detected clusters by eRosita and $\sim$ 400 SZ detected clusters by ACT, each of which are more than enough to produce an LSST mass calibration. Given that the previous ACT calibration uses a similar number of clusters [22], it is possible to produce competitive results using this early data alone. Constraining cosmology from calibration constitutes another project that would require estimating a covariance, modeling shear profiles (either hierarchically or stacked), redshift distribution estimation, and running cosmology chains. As an active member in LSST DESC, I am excited to contribute my skills to these efforts with my skills and help build a tomographic understanding of the cosmos.

This is an extremely exciting time to work in cosmology, and I have been at the forefront working on some of the first images taken by Rubin. I want to continue working with galaxy clusters and use them to set competitive constraints on cosmology which is only possible if we have a thorough handle on weak lensing systematics. Understanding the biases present and mitigating them is key to probing the Universe tomographically.

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